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Efficacy of historical context and exogenous features on deep learning for cooling load forecasting in chilled water plants

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Accurate day-ahead cooling load forecasting is a time series forecasting problem that is essential for optimizing the scheduling and energy efficiency of chilled water plants (CWPs). While previous studies have explored different forecasting models and input features, the combined effects of key factors such as historical context length (including regressor lags), sampling resolution, and exogenous inputs remain insufficiently examined for real industrial data. This study examines, under varying regressors and data granularity, the amount of historical context that provides stable and reliable forecasts with low computational cost. The experimental results show that (i) weekly scale look-back windows (LBW) provide an optimal balance between accuracy and computational cost for day-ahead horizons, (ii) hourly inputs often match or exceed finer-resolution accuracy and reveal distinct compute scaling: deep-learning (DL) models remain comparatively efficient as resolution increases, whereas the machine learning (ML) model, XGBoost, grows steeply, (iii) incorporating weather and calendar features enhances prediction accuracy around weekends and holidays. The DL model NHiTS, when integrating exogenous features and a 7-day LBW, improves forecasting accuracy by 50.8%, outperforming all other deep learning, machine learning, and statistical forecasting models by achieving a MASE of 1.88, compared to 3.82 for the baseline. These findings provide practical insights for smart-building and industrial applications, guiding practitioners such as building-energy engineers and facility managers in selecting optimal historical data lengths and parameter combinations to improve forecasting accuracy.

Keywords HVAC system, Time series forecasting, Cooling load, Machine learning, Deep learning, Look-back window

Abbreviations

AI	Artificial intelligence
ARIMA	Auto-regressive integrated moving average
BMS	Building management system
CWP	Chilled water plant
DBT	Dry-bulb temperature
DL	Deep learning
GPU	Graphics processing unit
HVAC	Heating, ventilation, and air conditioning
LBW	Look-back window
LSTM	Long short-term memory
MASE	Mean absolute scaled error
ML	Machine learning
N-HiTS	Neural hierarchical interpolation for time series forecasting
RMSE	Root mean squared error
TCN	Temporal convolutional network
TFT	Temporal fusion transformer
TR	Time resolution

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WBT	Wet-bulb temperature
XGB	Extreme gradient boosting
kWr	Kilowatts refrigeration
Std	Standard deviation

Heating, Ventilation, and Air Conditioning (HVAC) systems account for approximately 40–50% of the total annual energy consumption in buildings, with Chilled Water Plants (CWPs) alone contributing around 25–50% of this usage[1,2]. Therefore, improving the energy efficiency of CWPs is critical for reducing overall building energy consumption. To achieve efficient operation, it requires precise control strategies[3,4] and effective demand-side management[5,6], both of which depend on accurate time series forecasting of cooling loads. In practice, cooling load exhibits complex temporal dynamics in time series data that vary over weekdays, weekends, and public holidays[7]. The cooling load is also sensitive to external weather conditions and occasional operational anomalies, which produce nonstationary and nonlinear dynamics that challenge standard forecasting approaches[8,9] such as ARIMA (Auto-regressive Integrated Moving Average)[10,11]. To effectively capture these intricate temporal dynamics for day-ahead cooling load forecasting, it is crucial to determine the appropriate length of historical context or look-back window (LBW), which enables AI models to learn meaningful temporal patterns.

While previous studies[4,12–15] have focused on the architectural design of forecasting models for better capturing nonlinear patterns, several critical data-driven practical aspects remain insufficiently examined in a unified framework. In particular, the optimal length of historical context, or LBW, required to maximize forecasting performance on varying data sampling resolutions has not been consistently investigated. Without sufficient historical time series data (*i.e.* length of LBW), even advanced feature extraction strategies may fail, particularly in real-world industrial contexts where limited historical evidence can reduce forecasting reliability. Furthermore, while prior studies have explored the use of exogenous features such as calendar events and weather conditions[4,12], fewer works have examined their combined effects across different data granularities and forecasting configurations to ensure stable model performance. In addition, practical considerations such as the computational cost associated with different input configurations and data resolutions remain important but are not consistently addressed in the literature[16].

Recent review studies on building energy prediction highlight that forecasting performance depends on both model architecture and input configurations, such as historical context length, temporal resolution, and feature selection. These reviews also note that differences in datasets, forecasting horizons, and evaluation strategies make it difficult to establish consistent guidelines for practical deployment, particularly in real-world industrial settings[17–20].

Research questions. We aim to address the current limitations of cooling load forecasting approaches using real industrial CWP data. The study was guided by three research questions:

- How does the length of the LBW influence the performance of day-ahead cooling load forecasting models with different temporal resolutions?
- Where do state-of-the-art Artificial Intelligence (AI) models excel and fall short in capturing dynamic behaviors, operational cycles and anomalies in real-world cooling load data?
- How does the integration of exogenous features (e.g., calendar feature, dry bulb and wet bulb temperatures) influence the forecasting performance against baseline models?

This study addressed the research questions by conducting a systematic data-driven analysis of real industrial CWP data. The investigation was model-agnostic and aimed to investigate how the LBW configuration, data sampling resolution, and integration of exogenous features influence day-ahead forecasting accuracy and computational efficiency. The evaluation included multiple forecasting models, spanning statistical, ML, and DL approaches. These models were evaluated in three temporal resolutions (2.5-minutes, 5-minutes and hourly) for a day-ahead forecasting. Each model was tested in multiple LBW configurations with and without weather and calendar inputs.

The following are the contributions of the study.

- We present a model-agnostic framework evaluating the influence of LBW, temporal resolution, and exogenous inputs on real industrial CWP data for cooling load forecast, using diverse statistical (ARIMA), ML (XGB), and DL models (LSTM, TCN, TFT, and N-HiTS).
- We present the first systematic evidence that a weekly LBW (7 days) offers a strong accuracy–compute trade-off for day-ahead CWP forecasting, findings consistent across all evaluated model architectures.
- Hourly aggregation achieves accuracy comparable to or better than finer-grained inputs for operational use while substantially reducing training time and model complexity.
- We quantify the effect of exogenous features (calendar, dry-bulb, wet-bulb) and demonstrate their importance around weekends and public holidays, establishing robust, practitioner-oriented configuration rules for efficient and reliable forecasting pipelines.

Related works

AI methods for cooling load forecasting have evolved significantly, with both ML and DL approaches achieving notable success. However, most prior studies[14,21,22] have relied on fixed LBW configurations, without exploring how varying LBW sizes impact cooling load forecasting performance. In general, statistical models, such as ARIMA (Auto-regressive Integrated Moving Average)[10,11], have been widely used for cooling load forecasting, but they are limited in handling nonstationarity and complex temporal dynamics data[23,24]. More recently, ML methods, such as extreme gradient boosting (XGB)[25], have shown improved short- to medium-term forecasting

performance by capturing nonlinear relationships among input attributes[7,26]. Yet, ML struggles with long-range dependencies and high-dimensional feature spaces[27]. DL architectures, including Long Short-Term Memory (LSTM) networks[28], Temporal Convolutional Networks (TCN)[29], and Transformer-based models[30], have demonstrated a superior ability to model complex temporal structures in building energy systems[15,22,31,32]. Despite these advances, emerging models such as N-HiTS (Neural Hierarchical Interpolation for Time Series Forecasting)[27] remain underexplored for cooling load forecasting in CWPs, limiting the potential to achieve further predictive improvements.

Wang et al.[31] utilized 1-day LBW to forecast cooling loads for 1, 8, 16 and 24 hours ahead by incorporating XGB. Similarly, Yu et al.[7] incorporated XGB along with LBWs of 1 day and 2 days for HVAC heating and cooling load prediction. Chen et al.[12] used an ensemble ML algorithm with a 7-day LBW to forecast the next hour's cooling demand. DL methods, such as LSTM, TCN, and Temporal Fusion Transformer (TFT), have demonstrated superior performance in capturing complex temporal dependencies compared to statistical and ML methods. Notably, He et al.[4] applied LSTM for single-step (hour-ahead) chiller cooling load forecasting using a 1-day LBW. Similarly, Song et al.[14] proposed a hybrid model combining LSTM, TCN, and attention mechanisms (transformers) to predict short-term cooling load (1, 6, and 12 hours ahead), again using a 1-day LBW. Similarly, Xiao et al.[21] proposed a hybrid DL model using 1-day LBW to forecast the next 1 hour. Recent studies have also used Transformer models for improved sequential learning. Yu et al.[15] used a transformer-based approach with a 7-day LBW for day-ahead load forecasting. Li et al.[22] developed a sequence-to-sequence transformer model to predict 1 to 12 hours ahead using a 1-day LBW.

Generally, it can be observed that the length of the LBW in prior studies mostly falls between 1 to 7 days. However, very few studies selected the critical forecasting parameter, such as LBW, through a proper exhaustive search or analysis in a wide range of options with respect to AI model performance. Yet, the LBW is one of the most important parameters, as it determines how much historical context is required or sufficient for the model to effectively learn and predict future cooling load demands. In real-world industrial CWPs settings, such analysis can be a crucial factor and warrants further investigation. Additionally, while prior studies have explored the use of exogenous features, fewer works systematically evaluate their combined impact with LBW and data resolution on both forecasting accuracy and computational efficiency in CWP systems[4].

In addition, large-scale benchmarking studies such as the ASHRAE Great Energy Predictor III competition[33] have demonstrated that state-of-the-art performance in building energy prediction is often achieved using ensemble-based approaches dominated by gradient boosting tree models (e.g., XGBoost). Importantly, these studies highlight that data preprocessing and feature engineering play a critical role in determining model performance, sometimes exceeding the impact of model selection itself. Recent works have also explored advanced data-driven methodologies for energy forecasting. For instance, Zhang et al.[34] proposed a GPT-based automated framework that integrates feature engineering, model selection, and hyperparameter optimization, demonstrating the potential of transformer-based systems for improving both prediction accuracy and automation in energy forecasting pipelines. In parallel, Zhang et al.[35] investigated the impact of dataset partitioning strategies on regional cooling demand prediction, showing that interpolation-based data splits can significantly enhance model generalization and predictive accuracy under limited data conditions. This highlights the importance of data structuring and training strategies in time series forecasting tasks. Furthermore, comparative studies such as Runge et al.[36] have systematically evaluated machine learning and deep learning models for energy demand forecasting, demonstrating that models such as LSTM and XGBoost consistently achieve strong performance across different forecasting horizons, while also revealing trade-offs between prediction accuracy and computational efficiency.

These studies collectively reinforce that forecasting performance depends not only on model architecture, but also on data representation, feature engineering, and training strategies. However, they do not explicitly examine how key temporal factors such as look-back window length, data resolution, and exogenous feature integration jointly influence forecasting performance, which remains the primary focus of this study.

To address these research gaps, our study compared several state-of-the-art forecasting models with different LBW configurations, exogenous features, and temporal resolutions using real-world industrial CWPs data. This allowed for a systematic understanding of the trade-offs between input history length, temporal granularity, model architecture, and prediction accuracy.

Methods

This section outlines the methodological workflow developed to tackle several identified challenges. These challenges include identifying a model that is capable of capturing non-stationary and highly fluctuating load dynamics across weekdays, weekends, and public holidays. Additionally, the workflow addresses the selection of an appropriate LBW and temporal resolution to balance accuracy with computational cost. It also quantifies the impact of exogenous factors, such as weather and calendar features, on the accuracy of day-ahead cooling load forecasting. To evaluate how model architecture, LBW length, temporal resolution, and external variables collectively affect forecasting performance, various models were implemented and optimized. These models include ARIMA, XGB, LSTM, TCN, TFT, and N-HiTS. The overall workflow of this approach is illustrated in Figure 1.

Problem formulation

Let $\mathbb{D} = \{(x_t, y_t)\}_{t=1}^{n}$ denote the dataset, where $x_t \in \mathbb{R}^d$ represents the input feature vector at time t (including historical cooling load and exogenous features), and $y_t \in \mathbb{R}$ is the univariate target value representing the future system cooling load. The objective is to learn a forecasting function $f : \mathbb{R}^{l \times d} \rightarrow \mathbb{R}^h$ that maps a historical cooling load of length l into a prediction of future cooling load at h time steps. Specifically, for day-ahead forecasting (*e.g.*, $h = 288$ steps at 5-minute intervals), we seek to predict

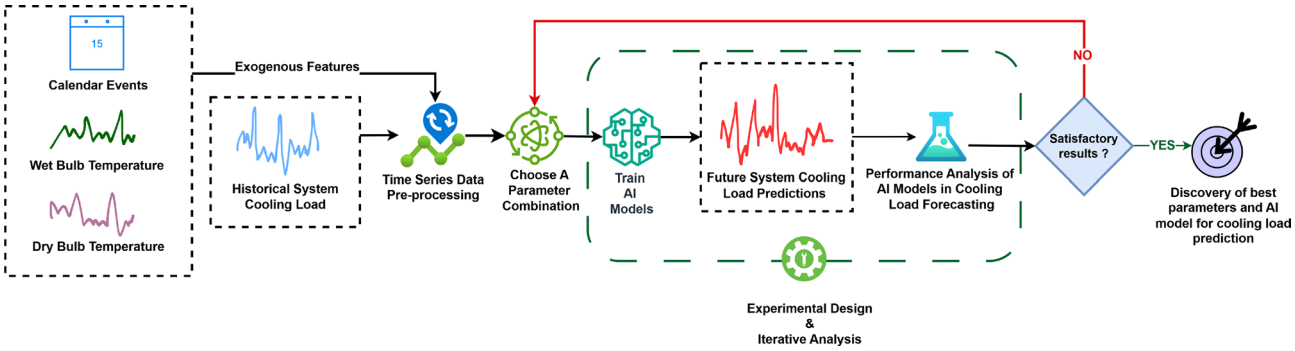


Fig. 1. The workflow of the forecasting processes in this work.

$$\hat{Y}_t = f(X_{t-l+1:t}), \tag{1}$$

where $X_{t-l+1:t} = (x_{t-l+1}, \ldots, x_t) \in \mathbb{R}^{l \times d}$ is the sequence of past time series values, and $\hat{Y}_t \in \mathbb{R}^h$ is the predicted univariate output series (future cooling load). Here, d denotes the input dimensionality determined by the number of exogenous features.

System and data description

The dataset used in this study was obtained from a physical CWP operating within a large commercial building in a tropical Southeast Asian city. The building represents a typical commercial facility equipped with a centralized cooling system. The plant consists of four chillers operating under a Building Management System (BMS), which monitors key operational metrics such as system cooling load and energy consumption. Data were recorded at a 5-minute frequency from April 1 to May 3, 2023, capturing approximately 8,640 time steps. The unit of system cooling load used in this study is kilowatts refrigeration (kWr), which represents the rate of heat removal by the CWPs. In general, cooling load or demand follows operational patterns, where the demand is relatively higher on weekdays and lower on weekends and public holidays. These patterns are critical for training forecasting models to learn the underlying operational cycles of the plant.

This study focuses on performing day-ahead cooling load forecasting, which involves predicting the system's cooling load for the next 24 hours using a certain amount of historical cooling demand data. The forecast horizons or number of time steps to predict vary depending on the data resolution. In our settings, there are 576 time steps for 2.5 minutes, 288 time steps for 5 minutes, and 24 time steps for 1 hour, suitable for long, medium, and short-term time series cooling load forecasting, respectively.

Exogenous features

To improve the accuracy and contextual awareness of day-ahead cooling load forecasting, we integrated three key exogenous features into the model inputs. They are:

- ***Calendar Feature***. This feature helps the models distinguish between operational and non-operational days, which are important for capturing the peak load on weekdays and zero-load behavior observed during weekends and public holidays.
- ***Dry Bulb Temperature***. It represents the ambient air temperature and is directly related to cooling demand, especially on hot weekdays.
- ***Wet Bulb Temperature***. It encompasses both temperature and humidity, refining the estimation of cooling load by accounting for latent heat effects that influence HVAC system performance.

These exogenous features were selected due to their strong physical and operational relevance to the behavior of CWPs.

Data preprocessing

The accuracy of the forecasting of time series cooling loads depends on the completeness and consistency of the data. However, building management system (BMS) datasets often include missing values due to sensor failures or connectivity interruptions. In our dataset, missing values were observed for two consecutive days (5th–6th April 2023), corresponding to approximately 6.7% of the total time series. These missing values were limited to specific periods rather than being randomly distributed and were addressed through a structured imputation strategy. To address this, we adopted a feature-specific imputation strategy designed to preserve the temporal and operational integrity of the dataset. For System Cooling Load, we applied weekday profile averaging. The imputed System Cooling Load $y^{\text{imp}}(t)$ at time t is computed as

$$y^{\text{imp}}(t) = \frac{1}{N} \sum_{i=1}^{N} y_i(t), \tag{2}$$

where $y_i(t)$ is the observed value at time t on the ith available weekday (excluding public holidays), and N is the number of such weekdays. For weather-related features: *Dry Bulb Temperature* and *Wet Bulb Temperature*, missing values were imputed by averaging readings from all available days to preserve realistic daily patterns. The imputed temperature $\hat{T}(t)$ at time t is computed as

$$\hat{T}(t) = \frac{1}{M} \sum_{j=1}^{M} T_j(t), \tag{3}$$

where $T_j(t)$ is the observed temperature at time t on the jth day, and M is the number of available days with valid temperature readings. Additionally, we introduced a categorical *Calendar Feature* $D(t)$, defined as

$$D(t) = \begin{cases} 1, & \text{if } t \text{ is a public holiday} \\ 2, & \text{if } t \text{ falls on a weekend (Saturday–Sunday)} \\ 3, & \text{otherwise (Weekday).} \end{cases} \tag{4}$$

Forecasting processes

The methodological workflow is to assess the influence of key forecasting parameters on the performance of the AI model. The key forecasting parameters were: LBW length (l), data sampling resolution (r), and the inclusion of exogenous features (x_t^{exo}). We defined the parameter pools as follows. The *Model Set* is given by $\mathscr{M} = \{\text{ARIMA}, \text{XGB}, \text{LSTM}, \text{TCN}, \text{TFT}, \text{N-HiTS}\}$, encompassing a diverse range of statistical, ML, and DL paradigms to capture varying temporal dynamics and feature dependencies. The *Resolution Set* is defined as $\mathscr{R} = \{2.5\text{-min}, 5\text{-min}, 1\text{-hour}\}$, allowing the investigation of how input data granularity affects model learning and forecasting performance. Finally, the *LBW* is specified as $\mathscr{L} = \{1, 2, 7, 8, 10, 14\}$ days, enabling exploration of different historical contexts to evaluate their impact on predictive accuracy.

The workflow of the forecasting processes and the sliding window approach utilized in this work are illustrated in Figures 1 and 2, and detailed in Algorithm 1. It begins with the input dataset $\mathbb{D} = \{(x_t, y_t)\}_{t=1}^{n}$, where y_t denotes the system cooling load at time t and $x_t \in \mathbb{R}^d$ is the associated feature vector containing historical load and exogenous variables such as weather conditions and calendar features. The data undergoes feature-specific preprocessing to impute missing values and encode categorical attributes such as the calendar feature $D(t)$. A configuration is then selected from the defined parameter pools: model set $\mathscr{M}$, resolution set $\mathscr{R}$, and look-back window set $\mathscr{L}$. For each combination $(l, r) \in \mathscr{L} \times \mathscr{R}$, sliding window datasets $\mathbb{X}_l^r$ and $\mathbb{Y}_l^r$ are generated, where each historical time series of length l is mapped to a prediction horizon of h steps. Each forecasting model $M_i \in \mathscr{M}$ learns a function $f : \mathbb{R}^{l \times d} \to \mathbb{R}^h$ that maps $X_{t-l+1:t}$ to the forecast $\hat{Y}_t$. Evaluation is performed using defined evaluation metrics. The approach ultimately identifies the optimal model M^* and parameter configuration that yield the most accurate day-ahead cooling load forecasts while considering computational complexity trade-off (detailed in the Results section). For benchmarking purposes, we incorporated seven state-of-the-art forecasting models, which included statistical, ML, and DL algorithms. These are:

- **ARIMA**[10]: A model traditionally used for univariate time series forecasting. ARIMA captures linear trends and seasonality under the assumption of stationarity but may struggle with complex nonlinear dependencies and sudden pattern changes.
- **XGB**[25]: An ensemble-based ML algorithm that constructs additive decision tree models via gradient boosting. XGB excels in short-term, structured pattern recognition but lacks temporal memory, making it less effective for capturing sequential dependencies.
- **LSTM**[28]: A type of recurrent neural network (RNN) architecture that incorporates memory cells and gating mechanisms, allowing it to model long-range temporal dependencies effectively. It is particularly well-suited for capturing complex sequential dynamics in non-stationary time series.
- **TCN**[29]: A fully convolutional DL architecture that replaces recurrent structures with causal dilated convolutions, offering large receptive fields for efficient long-horizon sequence modeling while maintaining stable gradient propagation.

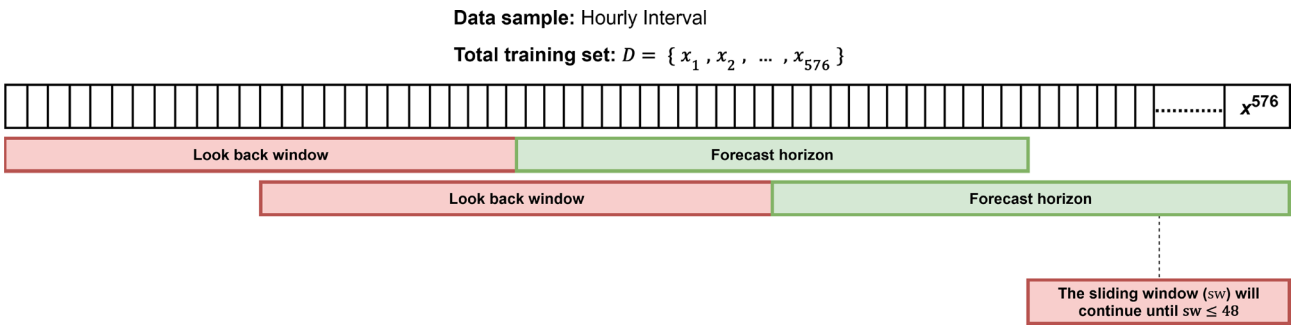


Fig. 2. Example of sliding window approach for hourly sampled data.

Input: Time series dataset $\mathbb{D}$ with t-minute resolution, including system cooling load, dry/wet bulb temperatures, and calendar feature.

Impute missing operational variables using weekday-profile averaging.
Impute missing weather variables using daily pattern averaging.
Label encode calendar feature $\mathbb{C}$ incorporating (4).
for each look-back window $l \in \mathscr{L}$ **do**
  for each sampling resolution $r \in \mathscr{R}$ **do**
    Construct sliding window datasets $\mathbb{X}_l^r$, $\mathbb{Y}_l^r$.
    Chronological train-test split.
  end for
end for
for each model $M_i \in \mathscr{M}$ **do**
  Optimize hyperparameters.
  Train M_i on $\mathbb{X}_{\text{train}}$.
  Generate predictions $\hat{Y}_t = M_i(\mathbb{X}_{\text{test}})$.
  Evaluate forecasts using standard performance evaluation metrics.
end for
Identify the optimal model M^* trained with best parameter combination based on minimum errors and maximum training efficiency.

Output: Day-ahead forecasts $\hat{y}_t$

Algorithm 1. Day-Ahead Cooling Load Forecasting.

- **TFT**[30]: A hybrid DL framework combining recurrent layers with multi-head attention mechanisms. TFT leverages both static covariates and temporal dynamics, providing interpretability through variable selection and attention-based long-term dependency modeling.
- **N-HiTS**[27]: A state-of-the-art DL model employing hierarchical basis interpolation and residual multi-resolution learning. N-HiTS is specifically designed for high-accuracy forecasting over multiple temporal scales, excelling at capturing both global trends and local fluctuations.

Hyperparameter configuration and selection

The performance of forecasting models is sensitive to the choice of hyperparameters. To ensure a fair and consistent comparison, we adopted a structured and controlled hyperparameter selection strategy across all models.

All DL models were implemented using the NeuralForecast library, while ML and statistical models were implemented using the MLForecast and StatForecast libraries from the Nixtla (also Nixtlaverse (https://nixtlaverse.nixtla.io/.)) ecosystem. To maintain a consistent training budget, the maximum number of training steps was fixed at 500 for all DL models. Early stopping was applied where applicable to reduce overfitting and unnecessary computation.

The input size for all models was defined based on the LBW. Rather than fixing the input length as a function of the forecast horizon, multiple LBW configurations were systematically evaluated, including 1 day, 2 days, 7 days, 8 days, 10 days, and 14 days. This design enables a structured investigation of the impact of historical context length on forecasting accuracy and model behavior across different temporal resolutions. Hyperparameter tuning was performed using a Bayesian optimization strategy within the Ray Tune framework[37]. For each forecasting model, 10 optimization trials were conducted. The search focused on key parameters such as learning rate, hidden dimensions, and architectural configurations, while keeping the search space limited to ensure computational feasibility and fair comparison. The final configurations were selected based on validation performance within the training set.

A summary of the final hyperparameter configurations for all models is provided in Table 1. This unified setup ensures that performance differences across models are primarily driven by model capability and input configuration, rather than differences in training budget or tuning effort.

Results

This section presents a detailed evaluation of forecasting models with varying LBW, exogenous feature configurations, and sampling rates. All simulations were conducted on a machine equipped with a 12th Gen Intel Core i7-12700H CPU (2.30 GHz), 32 GB RAM, and an NVIDIA GeForce RTX 3060 GPU. For the implementation of forecasting models, the Nixtlaverse's (https://nixtlaverse.nixtla.io/.) NeuralForecast, MLForecast, and StatForecast libraries, built on PyTorch, were utilized. To ensure the reproducibility of the primary experiments, the random seed for all models was initially set to 1, while multiple independent seeds were utilized later for the robustness analysis.

Model	Tuned Parameters (Search Space)	Fixed Parameters
TCN	Learning rate $\in \{5 \times 10^{-4}, 10^{-3}\}$; Encoder hidden size $\in \{16, 64\}$; Decoder hidden size $\in \{16, 128\}$; Kernel size $\in \{3, 5\}$	Max steps = 500; Dilations = [1,2,4,8]; Decoder layers = 2
N-HiTS	Learning rate $\in \{5 \times 10^{-4}, 10^{-3}\}$; Pool kernel size $\in \{[2, 2, 1], [16, 8, 1]\}$; Frequency downsampling $\in \{[2, 1, 1], [4, 2, 1]\}$	Max steps = 500; MLP units = [[512,512],[512,512],[512,512]]; Scaler = Robust
TFT	Learning rate $\in \{10^{-3}, 5 \times 10^{-3}\}$; Attention heads $\in \{2, 4\}$; Hidden size $\in \{20, 32\}$	Max steps = 500; Scaler = Robust
LSTM	Learning rate $\in \{5 \times 10^{-4}, 10^{-3}\}$; Encoder layers $\in \{2, 4\}$; Encoder hidden size $\in \{64, 128\}$	Max steps = 500; Context size = 10; Decoder hidden size = 64; Decoder layers = 2
XGB	Learning rate $\in \{10^{-3}, 10^{-2}\}$; Number of estimators $\in \{10, 100\}$	Max depth = 6; booster = gbtree

Table 1. Hyperparameter search space and fixed settings used during Bayesian optimization.

		LBW					
	Variant	1 d	2 d	7 d	8 d	10 d	14 d
TCN	baseline	5.83	6.50	4.08	3.89	4.42	3.95
	exogenous	6.74	9.21	2.43	2.42	2.99	2.24
LSTM	baseline	7.10	10.11	4.51	4.37	4.78	3.77
	exogenous	7.03	7.93	3.82	3.73	4.80	3.32
TFT	baseline	6.12	6.02	4.97	5.94	4.46	5.43
	exogenous	6.05	5.83	3.08	4.00	3.56	3.92
N-HiTS	baseline	4.00	4.08	3.82	4.25	2.61	3.41
	exogenous	4.47	4.47	**1.88**	2.52	2.51	3.08
XGB	baseline	3.94	3.39	3.09	3.04	3.01	3.13
	exogenous	4.20	2.45	3.11	3.06	3.04	3.11

Table 2. Comparative impact of exogenous features integration with AI algorithms against baseline versions for 1-hour resolution through MASE. The best result is highlighted with **bold** formatting.

Performance evaluation metrics

We evaluated the forecasting performance using two standard metrics: Root Mean Squared Error (RMSE) and Mean Absolute Scaled Error (MASE)[38,39]. This combination of metrics offers a balanced view of model performance: RMSE provides insight into the raw magnitude of forecast errors, while MASE contextualizes accuracy by benchmarking against a simple reference. Accordingly, Table 3 presents a detailed comparison of RMSE values among models, LBW lengths, and resolutions, while Table 2 focuses on MASE to evaluate the relative impact of incorporating exogenous features. This multi-metric approach supports a systematic and fair assessment of both model accuracy and robustness over diverse operational scenarios. Mathematically, let y_t denote the actual (observed) value at time t, $\hat{y}_t$ the predicted (forecast) value at time t, and n the total number of observations. The metrics are defined as

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}\left(\hat{y}_t - y_t\right)^2}, \text{and} \tag{5}$$

$$\mathrm{MASE} = \frac{\frac{1}{n}\sum_{t=1}^{n}\left|\hat{y}_t - y_t\right|}{\frac{1}{n-s}\sum_{t=s+1}^{n}\left|y_t - y_{t-s}\right|}. \tag{6}$$

In (6), assuming a seasonal period of s (where $s = 1$ for non-seasonal data), the MASE is computed by comparing the model's forecast errors with those from a simple seasonal naive model.

Look-back window impact

The choice of LBW size influences model performance, with most algorithms showing improved forecasting accuracy as the window length increases, as shown in the table. 2 and 3. The DL models, including N-HiTS, TCN, and LSTM, demonstrate performance gains with longer LBWs. For instance, TCN's MASE drops sharply from 9.21 (2-day LBW) to 2.43 (7-day LBW). At the same time, LSTM improves from 7.93 to 3.82, demonstrating its strong dependence on extended temporal information to capture weekly operational cycles and nonlinear dynamics. Notably, N-HiTS consistently achieves the lowest MASE in all window sizes, with optimal performance observed at a 7-day LBW (MASE is 1.88), indicating its strong ability to leverage extended temporal dependencies. The good results associated with the 7-day LBW may be due to its capability in capturing one full operational cycle, including weekdays, weekends, and their transitions. This allows the model to observe recurring patterns of start-up and shutdown behavior, occupancy-driven load shifts, and weather–load interactions over a full week,

		LBW					
	TR	1 d	2 d	7 d	8 d	10 d	14 d
TCN	2.5 m	885.60	1098.78	433.57	435.79	663.92	424.70
	5 m	851.27	1024.22	429.12	439.49	672.77	423.45
	1 h	1054.40	1171.40	338.50	338.67	477.85	324.27
LSTM	2.5 m	1047.37	1364.09	475.86	494.06	708.80	468.99
	5 m	864.55	1080.87	461.58	455.92	683.28	427.84
	1 h	1044.27	1060.98	439.19	432.16	591.11	401.46
TFT	2.5 m	664.18	1086.10	742.94	718.31	721.10	529.18
	5 m	813.64	913.04	456.58	548.74	536.68	526.16
	1 h	948.53	834.43	418.70	499.26	526.37	577.15
N-HiTS	2.5 m	664.19	829.35	469.62	512.38	406.75	443.99
	5 m	838.51	734.25	473.94	508.80	400.06	465.43
	1 h	587.00	586.96	**264.91**	344.71	381.82	503.94
XGB	2.5 m	381.40	544.02	464.70	444.94	621.09	622.12
	5 m	383.86	544.34	462.82	440.90	615.35	622.13
	1 h	678.33	492.36	603.71	603.17	596.18	604.39
ARIMA	2.5 m	863.65	863.75	863.72	863.36	863.44	863.22
	5 m	862.72	862.76	863.21	863.21	863.53	863.07
	1 h	853.60	852.90	852.10	851.95	852.09	852.01

Table 3. Systematic benchmarking of performance (RMSE) for the optimized version of AI algorithms for cooling load (kWr) forecasting at 3 different time resolutions (TR) of data in the real-world industrial dataset from a CWPs. The best result is highlighted with **bold** formatting.

providing sufficient contextual information for day-ahead prediction while remaining short enough to avoid overfitting and excess compute.

XGB shows competitive performance by achieving the lowest MASE of 2.45 at a 2-day window, but shows diminishing returns beyond 7 days and suffers from limited generalization to unseen conditions such as public holidays. In contrast, N-HiTS outperforms XGB at 7-day LBW on coarser resolutions (e.g., 1-hour, where it achieves 264.91 RMSE vs. 603.71 for XGB). From this analysis, we found that XGB excels in short-term temporal reasoning at high resolution for cooling load forecasting. However, N-HiTS is superior for long-term temporal modeling and generalization. TFT maintains a relatively consistent performance in various LBW combinations but does not outperform N-HiTS. In contrast, ARIMA shows a minimal variation in all LBW sizes, with nearly no change in MASE (approximately 5.47). This observation suggests that linear statistical models are unable to effectively exploit additional historical context, particularly in nonlinear and fluctuating demand environments. From a practical standpoint, a 7-day LBW appears optimal for balancing accuracy and training time. Though XGB performs well for short horizons, its training time escalates substantially with larger LBWs (*e.g.*, over 80 minutes at 8-day LBW and 5-minute resolution), unlike DL models such as TCN and N-HiTS, which maintain high performance with significantly lower computational time. Finally, these results highlight the importance of selecting an appropriate LBW, ideally one full operational week, to extract meaningful load patterns while avoiding overfitting and unnecessary computational complexity.

Baselines vs. exogenous feature

In our simulation, all models were initially trained using only historical cooling load data to establish baseline performance. We then incorporated the exogenous features and evaluated the resulting performance using MASE, as presented in Table 2. The results show substantial improvements in accuracy in most DL models. Notably, N-HiTS achieved the lowest overall MASE of 1.88 with a 7-day LBW, improving from its baseline MASE of 3.82. TCN and TFT also demonstrated significant gains, with TCN improving from 4.08 to 2.43, and TFT from 4.97 to 3.08, under the same configuration. These results highlight the deep models' ability to leverage exogenous context to better differentiate between typical and atypical load profiles. Notably, XGB showed only marginal improvements with the addition of exogenous features (e.g., MASE changed from 3.09 to 3.11 at 7-day LBW), suggesting that tree-based models may be less effective at extracting temporal dependencies from sequential auxiliary inputs. However, ARIMA can only accept univariate input and was trained solely on historical cooling load data, limiting its ability to learn additional information from exogenous factors such as calendar or weather data.

Overall, the incorporation of exogenous variables proved especially beneficial for DL architectures, enabling them to more effectively model non-linear transitions, such as load dips during holidays or temperature-driven peaks, thereby improving day-ahead forecast reliability.

In-depth performance analysis of AI models

The overall comparison of the forecasting models using a 7-day LBW reveals significant differences in their ability to capture temporal patterns, fluctuations, and holiday effects in cooling load dynamics, illustrated in

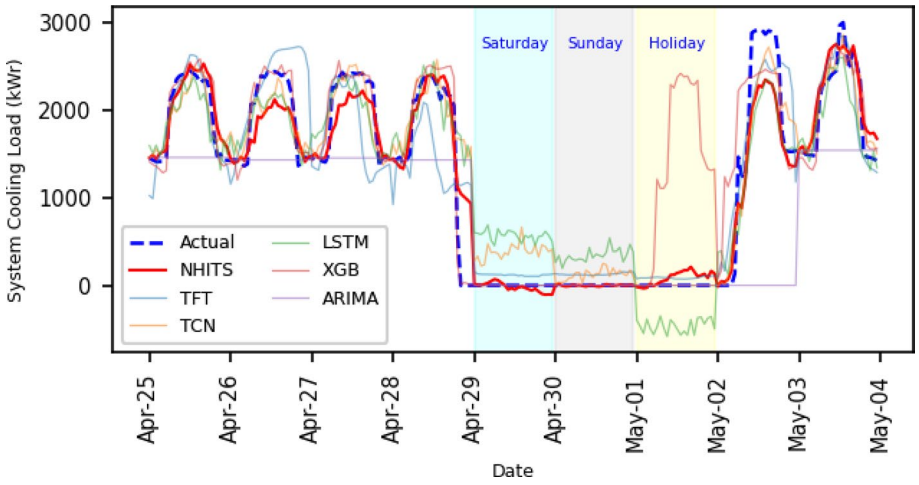


Fig. 3. Comparison of actual versus predicted day-ahead cooling load (kWr) using six forecasting algorithms with a 7-day look-back window on hourly sampled data.

	5 m	1 h	2.5 m	Avg.
TFT	382.5	31.4	413	275.63
LSTM	118.1	17.1	1977	704.07
TCN	103.7	17.6	1934	685.10
N-Hits	135	36.4	297	**156.13**
XGB	2927.9	19.8	33640	12195.9

Table 4. Computation time (s) required by each machine-learning algorithm at three sampling intervals–2.5 min, 5 min, and hourly–using a 7-day look-back window.

Fig. 3. N-HiTS consistently achieved the lowest overall errors–e.g., RMSE = 264.91 on hourly data with 7-day LBW (Table 3) and showed strong alignment with the actual load profile in all operating conditions, including weekdays, weekends and public holidays (Fig. 3). It also demonstrated robust performance among other time resolutions, maintaining relatively low errors with reasonable training times (e.g., 36.4 seconds on hourly data, 135 seconds on 5-minute data, shown in table 4).

TCN closely follows actual load patterns during high-demand weekday periods (e.g., April 26–28) and performs competitively, with RMSE = 338.5 on hourly data and RMSE = 429.12 on 5-minute data at a 7-day LBW. Additionally, TCN achieved the second-best performance at a 14-day LBW, with an RMSE of 324.27 on hourly interval data, demonstrating its strong ability to capture extended temporal patterns. However, it slightly underestimates the load on inactive days, such as weekends, resulting in over-prediction (Fig. 3).

XGB excels in modeling weekday patterns and weekend drop-offs, with a notably low RMSE of 383.86 at a 1-day LBW using 5-minute data. However, its accuracy does not improve significantly with larger LBWs (e.g., RMSE = 462.82 at 7 days, 615.35 at 10 days), and it fails to recognize anomalies such as public holidays, such as May 1, treating them as regular weekdays, which results in overestimation (Fig. 3). This observation suggests that XGB relies on structured historical trends without adaptive context, despite its strength in short-term learning.

TFT and LSTM follow general load trends well but occasionally exhibit instability or lag during sudden changes. At 7-day LBW with 5-minute data, LSTM reaches RMSE = 461.58, and TFT records RMSE = 456.58, showing stable yet slightly less precise performance than N-HiTS. ARIMA, in contrast, consistently struggles to model nonlinear behavior or abrupt transitions, producing flatter forecasts (Fig. 3). For instance, its RMSE remains largely static among LBWs, ranging narrowly between 862.72 and 863.53 with 5-minute data, further highlighting its limited capacity to capture dynamic changes.

Overall, most of the algorithms demonstrate higher error rates (RMSE) when using 5-minute data (*e.g.*, N-HiTS at 7 days: 473.94) and 2.5-minute data (*e.g.*, N-HiTS at 7 days: 469.62) compared to hourly data (264.91). The finer resolution of the data introduces more fluctuations and noise, making forecasting more challenging. Table 3 provides a detailed comparison of RMSE values among six LBW settings for each algorithm under both 5-minute and hourly resolutions.

Computational complexity analysis

To evaluate the robustness of each model, we tested three temporal resolutions: 2.5 minutes, 5 minutes, and 1 hour. The results and trade-offs are summarized in Table 4 for LBW of 7 days. The 1-hour resampled dataset provides the most efficient configuration, requiring only 24 time steps per day-ahead forecast. This substantially

reduces training time for all models, with all completing under 40 seconds (e.g., TCN: 17.6, TFT: 31.4, XGB: 19.8, N-HiTS: 36.4), as shown in Table 4, while still maintaining competitive accuracy.

Though the 5-minute interval data (288 steps per forecast) slightly improves prediction accuracy (from Table 3) for some DL models (e.g., TCN), this improvement comes at a notable computational cost. For instance, XGB requires over 48 minutes (2927.9 seconds) to train with this resolution, compared to just 2–6 minutes for DL models such as N-HiTS (135 seconds) and TFT (382.5 seconds). It is important to note that the computational behavior of XGB observed in this study differs from the commonly reported efficiency of tree-based models in the literature[33,36]. While XGB is generally considered computationally efficient, particularly for structured datasets with moderate feature sizes, its training cost increases significantly when the number of input features becomes large. In our experimental setup, the use of high-resolution data and extended look-back windows introduces a substantial number of lag-based features, leading to a high-dimensional input space. This increase in feature dimensionality results in higher computational overhead for tree construction and split evaluation, thereby reducing the efficiency advantage typically associated with XGB.

Upsampling the data to 2.5-minute intervals (576 steps per forecast horizon) introduces significant training overhead and diminishes predictive performance for most models. While N-HiTS and TFT remain relatively efficient at this scale (under 300 seconds and 413 seconds, respectively), models such as TCN and LSTM experience noticeable slowdowns (1934 seconds and 1977 seconds, respectively). XGB's training time exceeds 560 minutes (33640 seconds), making it impractical at this frequency. The added resolution also introduces potential overfitting or noise sensitivity, with no consistent gain in accuracy.

Overall, the 1-hour resolution offers the best balance of speed and accuracy, while 5-minute data may be justified when capturing finer temporal patterns is necessary. Excessively high-frequency data, such as 2.5-minute intervals, should be avoided unless required due to diminishing returns and computational inefficiency.

Optimal configuration summary

Based on systematic evaluations in all forecasting models, multiple LBW sizes, and three sampling resolutions, the optimal configuration for forecasting the day-ahead cooling load in CWPs is identified as the N-HiTS model with a 7-day LBW, 1-hour sampling rate, and inclusion of key exogenous characteristics (Day type, dry bulb, and wet bulb temperature). This setting achieved the best overall performance with the lowest RMSE (264.91) and MASE (1.8761), effectively capturing weekly operational cycles and external influences such as holidays and weather. While XGB showed strong accuracy at shorter LBWs, it suffered from a high computational cost, exceeding 2927.9 seconds for 5-minute data and 33640 seconds for 2.5-minute data. In contrast, N-HiTS offered competitive accuracy with the lowest average training time ($\sim$156 seconds) shown in Table 4. These results confirm that a balanced configuration of model architecture, temporal resolution, and historical context is critical for achieving accurate and scalable cooling load forecasting.

Interpretation of findings

The experimental results underscore several key insights into the behavior of forecasting models under varying parameter settings. First, deep learning models such as N-HiTS, TCN, and LSTM benefit significantly from longer LBW, with improved performance metrics when moving from short (1–2 days) to extended (7–10 days) windows. This outcome confirms the importance of historical context in capturing weekly patterns and operational cycles in cooling load dynamics. In contrast, ARIMA shows minor sensitivity to window size, reinforcing its limited capacity to model complex temporal dependencies and nonlinearities.

Second, the inclusion of exogenous features such as Day Type and Dry/Wet Bulb Temperatures markedly improves forecasting accuracy, particularly for models like N-HiTS and TFT. These features enable the models to adapt to abrupt load changes during weekends and public holidays, where baseline-only models typically underperform. The rigorous analysis also highlights the limitations of ensemble ML models such as XGB, which show strong performance in structured weekday patterns but fail to generalize well to unseen scenarios such as holidays.

Third, the impact of sampling resolution is non-trivial. While higher-frequency data (5 minutes and 2.5 minutes) offers more granular insights, it also introduces noise and significantly increases the computational burden. The 1-hour resolution provides a favorable trade-off between accuracy and efficiency, particularly for real-time or large-scale applications. N-HiTS emerged as the most robust model in all scenarios, demonstrating stable accuracy with low training time and adaptability to different resolutions and feature sets.

Finally, these findings highlight the importance of selecting the right combination of model type, input features, historical context, and temporal resolution to build scalable and accurate forecasting systems for energy-efficient operation of CWPs.

Ablation summary and reporting of relative improvements

For any error metric m (lower is better), we report the percentage change as

$$\%\Delta m = 100\,\frac{m_{\text{variant}} - m_{\text{baseline}}}{m_{\text{baseline}}} \tag{7}$$

So negative values indicate improvement. We also report the improvement factor $m_{\text{baseline}}/m_{\text{variant}}$ for interpretability.

Key ablation finding. Adding exogenous features (weather + calendar) to NHITS at a 7-day look-back reduced MASE by 50.8% (from 3.82 to 1.88), corresponding to a $\sim$2.03$\times$ improvement.

Model	LBW	Seed 0	Seed 1	Seed 7	Seed 42	Seed 1234	Mean	Std
N-HiTS	1 d	3.66	4.47	3.98	5.49	4.39	4.40	0.69
	7 d	**2.00**	**1.88**	**1.96**	**1.94**	**1.84**	**1.92**	**0.06**
	10 d	2.74	2.51	2.71	2.52	2.57	2.61	0.11
TCN	1 d	10.58	6.74	13.45	8.15	8.00	9.38	2.66
	7 d	2.23	2.43	2.48	2.28	2.71	2.43	0.19
	10 d	4.79	2.99	4.72	5.32	1.86	3.94	1.45
TFT	1 d	5.47	6.05	5.43	4.53	7.44	5.78	1.07
	7 d	2.78	3.08	2.55	3.65	3.30	3.07	0.43
	10 d	2.59	3.56	3.87	4.00	3.35	3.47	0.56
LSTM	1 d	7.10	7.03	6.53	7.22	7.48	7.07	0.35
	7 d	5.37	3.82	6.19	6.20	6.49	5.61	1.09
	10 d	13.56	4.80	8.41	11.02	8.12	9.18	3.30

Table 5. Robustness analysis of DL models using five random seeds (0, 1, 7, 42, and 1234) evaluated at the 1-hour sampling resolution with exogenous features.

Robustness analysis of deep learning models

To assess the robustness of DL models against random initialization, all experiments were repeated using five arbitrary random seeds (0, 1, 7, 42, and 1234). This analysis is important because DL models involve several stochastic components, including random weight initialization, data shuffling, and optimization dynamics, which can lead to variations in forecasting performance across different training runs. Table 5 reports the corresponding MASE values, together with the mean and standard deviation (Std), for selected look-back windows (1, 7, and 10 days). Overall, the relative performance ranking of the models remained largely consistent across different seeds, indicating that the observed performance trends are not dependent on a particular random initialization. Among the evaluated models, N-HiTS demonstrated the lowest variability across seeds, particularly at the optimal 7-day LBW (MASE = 1.92 ± 0.06), indicating strong robustness and stable forecasting performance. In contrast, TCN and LSTM exhibited greater sensitivity to random initialization under certain configurations, especially at shorter and longer LBWs.

Limitation

This study focuses on a single plant in a tropical climate using a relatively short dataset (approximately one month of data). As a result, the findings may not fully capture seasonal variations in cooling load behavior, and model performance may differ under other conditions, such as peak summer or shoulder seasons. Future work should generalize findings across different geographies, seasons, and building types using longer-term datasets.

Uncertainty quantification and hybrid model ensembles could also be explored. For a fair comparison benchmark, this study maintains standard hyperparameter configurations for deep learning models. However, deep learning models could be further hyper-tuned or extended with more complex architectures, which may improve performance when larger datasets (e.g., tens of thousands of samples) are available.Additionally, the results are influenced by the selected forecasting framework. This study adopts a sliding window approach for multi-step forecasting; alternative approaches (e.g., recursive or direct forecasting strategies) may yield different performance outcomes. In future studies, this could be further explored to better understand the behavior of models under different training strategies and data configurations.

Despite these limitations, the findings provide consistent and practically relevant insights for cooling load forecasting under real-world operational conditions.

Discussion

The results presented in this study provide several key insights into how model performance varies with input parameters and algorithm choice. These findings align with the fundamental nature of both the data and the AI models.

RQ1: Effect of look-back window (LBW) length. One of the most prominent findings is that extending the length of LBWs from 1 or 2 days to 7 days improved the performance of DL models such as N-HiTS, TCN, LSTM, and TFT. These models exhibited lower forecasting errors when provided with longer historical data. This improvement is likely due to the fact that cooling load patterns often follow weekly operational cycles, and shorter input windows are insufficient to capture these trends[14,21,22]. The 7-day LBW enabled the models to learn recurring patterns that occur throughout the weekdays and weekends more effectively, resulting in more accurate day-ahead predictions. The performance enhancements were particularly notable in N-HiTS and TCN, both of which employ hierarchical and convolutional architectures, respectively, architectures that are especially well-suited for capturing long-range temporal dependencies[40,41]. XGB, while competitive with shorter LBW, showed limited adaptability to sudden shifts such as public holidays (Fig. 3). This outcome is expected, as tree-based models tend to rely heavily on structured historical trends and struggle with context shifts[42]. Its superior short-term accuracy comes at a cost of high computational time.

Moreover, the analysis of sampling resolution revealed that while high-frequency data (*e.g.*, 2.5 minutes) offers more detail, it introduces noise and significantly increases training time (computational complexity). The

1-hour resolution, despite being coarser, often offered comparable accuracy with much faster training times, making it preferable for real-time or resource-constrained environments.

RQ2: Where state-of-the-art AI models excel and fall short on real CWP data. The comparative results reveal that sequence-based DL architectures–particularly N-HiTS and TCN-handle the nonlinear and highly fluctuating cooling-load patterns in real CWP data better than the alternatives. Their capacity to integrate multi-scale temporal dependencies enables them to capture both rapid weekday demand spikes and the gradual transitions into weekends and holidays, which traditional algorithms (ARIMA) often miss[43]. For example, N-HiTS achieves the most consistent alignment with actual load patterns across all operating conditions, while TCN effectively captures weekday peaks but tends to slightly overestimate loads during inactive periods. These outcomes suggest that hierarchical and convolutional mechanisms are inherently well-suited to learning long-range dependencies within complex operational cycles[44,45].

In contrast, ensemble tree-based methods such as XGB perform competitively for short-horizon or high-resolution forecasts but struggle to generalize when the input horizon is extended. Their reliance on fixed historical relationships limits adaptation to contextual shifts (e.g., holidays or unusual weather). As a result, although XGB is computationally efficient for short-term forecasts, its lack of sequence awareness reduces robustness under dynamic conditions. Similarly, traditional linear models such as ARIMA fail to capture the nonlinear transitions and operational variability inherent in chilled-water systems, leading to static or flattened forecasts.

Furthermore, the robustness analysis using five independent random seeds demonstrated that the observed performance trends were stable across different random initializations. Although some variability was expected due to the stochastic nature of deep learning training, the relative ranking of the models remained largely unchanged. Notably, N-HiTS consistently achieved the lowest mean MASE and the smallest standard deviation at the optimal 7-day LBW (MASE = 1.92 ± 0.06), indicating strong robustness and reliable convergence. In contrast, TCN and LSTM exhibited greater variability under certain configurations, suggesting higher sensitivity to random initialization. These results provide additional confidence that the superior performance of N-HiTS is attributable to its architecture and ability to capture temporal dependencies, rather than the outcome of a favorable random seed.

RQ3: Influence of exogenous features relative to baselines.

Exogenous features, such as calendar and weather-related factors, provide complementary context that the cooling load history alone cannot supply. Calendar indicators (weekday, weekend, public holiday) mark routine regime changes in plant operation, helping models avoid carrying weekday load levels into non-operational days and improving day-to-day stability around schedule shifts. Weather-related features encode physical drivers of demand: dry-bulb temperature reflects sensible cooling needs, while wet-bulb captures humidity effects linked to latent loads and plant performance. These signals summarize gradual meteorological trends and acute heat events that shift the baseline load.

The advantages of using exogenous features were especially noticeable in models that are designed for sequence-to-sequence learning, including N-HiTS, TCN, and TFT. These models were able to utilize the correlations among exogenous features to adjust for fluctuations caused by operational and non-operational days, as well as variations in weather. Empirically, adding weather and calendar features to N-HiTS with a 7-day LBW reduces MASE by ∼50.8% (3.82→1.88; ∼2.03×), consistent with this mechanism. In contrast, XGB demonstrated marginal improvements, highlighting its limited ability to capture sequential dependencies in auxiliary inputs (TABLE 2). Moreover, models that lack this contextual support (Exogenous features), such as ARIMA, struggled to produce meaningful nonlinear forecasts.

Overall, the findings highlight the importance of aligning the model architecture, historical input length, and feature design with the operational characteristics of the forecasting problem. The best-performing configurations were not only the most accurate but also computationally feasible, which is essential for practical deployment in building energy systems.

Conclusion

This study evaluates AI-based forecasting models for predicting day-ahead cooling loads in CWPs. It focuses on key parameters such as the length of the LBW, temporal resolution, and the integration of exogenous features. By systematically benchmarking state-of-the-art AI models with various LBWs and sampling rates, we show that model performance varies significantly based on input configuration. The results indicate that deep learning models such as N-HiTS and TCN benefit notably from longer LBWs of at least 7 days, as they can more effectively capture recurring weekly patterns in cooling load. Additionally, incorporating external variables, such as calendar features and wet bulb and dry bulb temperatures, further improves forecasting accuracy, especially for sequence-based models. For instance, N-HiTS achieved the lowest MASE of 1.88 (a mean MASE of 1.92 across multiple random seeds) when using 1-hour resolution data with exogenous features. In contrast, while XGB performs well with shorter LBWs and high-resolution inputs, it struggles to generalize across variable patterns, such as those observed on public holidays, and incurs higher computational costs.

In response to our research questions: (i) the length of historical context greatly affects day-ahead accuracy, with a weekly-scale LBW (7 days) providing the best accuracy–compute balance across models and input resolutions in our setting; (ii) among model families, sequence-based deep models, especially N-HiTS, were the most robust across configurations, while XGB remained competitive only at short LBWs and finer resolutions and generalized less reliably across regime changes; and (iii) adding exogenous information (calendar plus dry-/wet-bulb temperatures) consistently improved stability and reduced errors, particularly around weekends, public holidays, and heat events.

Overall, these findings underscore the importance of selecting the right combination of model type, input window (LBW), and feature sets to balance prediction accuracy and computational efficiency. The optimal

configuration involves using N-HiTS with a 7-day LBW and external features, resulting in lower average computational times, making it the most effective choice for real-world CWP forecasting. This work highlights the trade-offs between models' accuracy and complexity, providing practical guidance for facility managers and engineers who deploy AI-based solutions for predictive maintenance and energy-efficient operations. Future work will focus on expanding the dataset to capture seasonal variability and improve model generalizability in multiple plant sites. As DL models tend to perform better with more extensive and more diverse data, incorporating long-term operational records could further enhance prediction accuracy and robustness.

Data availability

The underlying datasets are proprietary, real-world industrial records owned by our industry partner, Exergenics Pty Ltd, and governed by binding confidentiality agreements; therefore, they are not publicly available. Access to the raw datasets may be considered on reasonable request and is contingent upon prior written approval by the data owner and execution of a confidentiality agreement.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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